Large Language Model–Driven Reinforcement Learning for Weekly Portfolio Allocation

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MPhil Thesis Defense

January 27, 2026

Roadmap

- **Motivation and problem setting**
  - What is portfolio allocation?
  - Why it matters in practice
  - What this thesis asks
- **Task formalisation and experiment setup**
  - Empirical instantiation and benchmarks
  - RL environment and PPO formulation
  - LLM analyst pipeline
  - Train / validation / test design
- **Key results**
  - Convergence and overall performance
  - PPO vs. EW and LLM integrations
- **Discussion**
  - Limitations, contributions and future work
- **Related work**
  - Textual analysis in finance and financial LLMs
  - Agentic LLM systems and FinRL competitions

Motivation and Problem Setting

What is Portfolio Allocation?

- **Portfolio management:** deciding how to distribute capital across a basket of assets over time.
- At each decision point:
  - Choose a weight vector $(w_1, \ldots, w_N)$ over assets (stocks, bonds, cash, etc.).
  - Weights represent the percentage of current total capital allocated to each asset.
- **Goal:** balance
  - long-run return, and
  - risk: volatility, drawdowns, tail events.
- This thesis focuses on:
  - 20 liquid U.S. large-cap stocks
  - weekly rebalancing decisions

Modern Portfolio Theory (Markowitz, Mean–Variance)

- 1950s: Harry Markowitz formalised portfolio choice.
- **Assumption:** future returns are uncertain, but their statistical structure can be modelled and optimised.
- **Key ideas:**
  - **Diversification reduces risk:** correlations between assets matter.
  - **Efficient frontier:** a mathematical tool to identify portfolios with the best risk–return trade-offs.

Portfolio Management in Practice

- **Who uses portfolio allocation?**
  - Pension funds, mutual funds, insurance companies.
  - Quantitative hedge funds, ETFs, and robo-advisors.
- **Scale:**
  - Each category manages assets on the order of *tens of trillions* of USD.
  - Together, they represent a large share of roughly $300T in global financial assets.
- **Operational reality:**
  - Portfolio allocation is a day-to-day routine: rebalancing, responding to shocks, managing constraints.
  - Even small improvements at similar risk levels can have large monetary impact.
- **Implication for this thesis:** portfolio allocation is a realistic and meaningful testbed for RL and LLM-driven signals.

Overarching question (High Level)

Core Question
Can large language models provide informative structured signals and risk scores from multiple textual sources (e.g., news, earnings-call transcripts, SEC filings) that help a reinforcement learning agent make better weekly portfolio allocation decisions than:

- classical allocation rules (equal-weight, minimum-variance), and
- a price-only RL agent?

Research Questions

- **RQ1: LLM features as observations**
  - How do different textual sources and feature extractors affect an RL agent's training and trading performance?
- **RQ2: LLM-based risk shaping**
  - Can LLM-derived risk scores meaningfully shape an RL agent's risk awareness?
- **RQ3: Training schedules and curricula**
  - How do different training mechanisms affect the RL agent's stability, behaviour, and robustness?

Task Formalisation and Experiment Setup

Empirical Task Instantiation

- Universe: 20 equities
  - 20 liquid large-cap U.S. equities (S&P500-like).
- Decision calendar:
  - 334 weekly rebalancing dates.
  - From April 13, 2019 to late August 2025.
- At each week:
  - Agent observes portfolio state, recent market data, and (optionally) LLM signals.
  - Chooses a weight vector over assets
  - Environment executes trades under transaction costs, holds the portfolio for one week, and returns a risk-aware scalar reward.

Classical Baselines

- **Equal-weight (EW) portfolio**
  - Rebalances the 20 equities to equal weights each week.
- **Minimum-variance (MV) portfolio**
  - Classical Markowitz / Modern Portfolio Theory benchmark.
  - Uses a 26-week rolling covariance matrix of equity returns.
  - Chooses weights that minimise portfolio variance subject to standard constraints, corresponding to the global minimum-variance portfolio on the Markowitz efficient frontier.
- **Common setup for comparability**
  - Same 20-asset universe and 100-week decision calendar as the RL agent.
  - Weekly rebalancing.
  - Same 10 bps proportional transaction cost on buys and sells.

RL Formulation as an MDP

- Weekly portfolio allocation $\Rightarrow$ Markov decision process.
- For each split (train / val / test):
  - Ordered list of weekly decision dates from Nasdaq calendar.
  - One episode = traverse this sequence once.
- At each week t:
  - Observe a structured feature.
  - Choose a new portfolio weight vector.
  - Apply transaction costs, hold for one week.
  - Receive scalar reward combining return and risk penalties.
- Architectural novelty concentrated in:
  - Observation design (feature extractor + LLM signals).
  - Risk-aware reward shaping.

Data Modalities and Observation

- Underlying data modalities:
  - Market prices and volumes (daily) $\Rightarrow$ technical indicators.
  - News bundles per (ticker, weekly): sparse and uneven over tickers and weeks.
  - Earnings-call transcripts (quarterly): made by company management.
  - SEC filings (quarterly and yearly): regulatory filings.
- Observation at week t includes:
  - Current portfolio weights and cash fraction.
  - Market features from prices over recent price history.
  - Optional LLM-derived signals and risk scores (RQ1/RQ2/RQ3).

Feature Extractor and PPO Agent

- **Feature extractor**
  - 1D Conv over a fixed history window of prices per stock (or an MLP in simpler variants).
  - Cross-asset aggregation MLP (or via pooling / attention layers).
  - Outputs a single fused feature vector for week t.
- **Why PPO?**
  - Supports continuous actions (weights over assets).
  - Clipped objective stabilises training in a data-scarce regime.
- **Actor–critic structure (from StableBaseline3)**
  - Actor: fully-connected net producing a logit vector; softmax $\Rightarrow$ portfolio weights over assets.
  - Critic: fully-connected net mapping the same features to a scalar value estimate for advantage computation.

PPO Objective and Risk-Aware Reward

- **PPO clipped objective**

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\,A_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,A_t\big)\Big],$$

  where $r_t(\theta)$ is the policy ratio and A_t is the advantage computed from discounted sums of per-step rewards.

- **Risk-aware reward at week t**

$$r_t = \lambda\Big(r_t^{\text{return}} - \beta\, z_t^{\text{mkt}}\, E_t - \gamma\, z_t^{\text{llm}}\Big).$$

- **Idea**
  - When $\beta = \gamma = 0$, r_t reduces to the log return $\log\big(1 + R_t^{\text{port}}\big)$.
  - Positive (β, γ) encourage the agent to (i) avoid accumulating downside risk in turbulent markets and (ii) moderate exposure to firms flagged as risky by the LLM analysts.

LLM Analyst Pipeline Overview

- Text sources differ in timescale and price impacts:
  - SEC filings: mandatory regulatory reports on a firm's financial condition and risks.
  - Earnings-call (EC) transcripts: quarterly calls where management discuss results and outlook.
  - News / technical commentary: faster, event-driven market coverage.
- All text is aligned to the weekly decision calendar:
  - For week t, only use documents with timestamps $< t$.
- Two-tier analyst architecture:
  - **Tier 1**: channel-specific analysts (SEC, EC, News, Technical).
  - **Tier 2**: senior analyst synthesising across channels per (ticker, week).
- Shared schema for RL:
  - `signal`: directional score (observation).
  - `risk_score`: downside risk intensity (reward shaping).
  - `confidence`: reliability in $[0, 1]$.
  - `rationale`: short free-text explanation (for humans).

Tier-1 and Tier-2 Analysts

- **Tier-1: channel analysts**
  - SEC analyst: 1–3 year structural view from 10-K / 10-Q (map–reduce).
  - EC analyst: 1–4 quarter view from earnings-call transcripts (map–reduce).
  - News analyst: weekly updates.
  - Technical analyst: commentary over price/volume patterns.
- **Tier-2: senior analyst**
  - Sees all Tier-1 digests per (ticker, week).
  - Debates, stress-tests, and reconciles across channels.
  - Emits a single schema-constrained record to RL.
- **Prompting and staleness awareness**
  - SEC / EC prompts emphasise structural fundamentals and staleness.
  - News prompts emphasise material 1–4 quarter updates, not noise.

Example: News Prompt and Rubric

News System Prompt (excerpt)

- *"You are a fundamental equity news analyst working for a systematic portfolio manager."*
- SEC/EC provide the slower baseline (1–3 years / 1–4 quarters).
- *"Your job is to decide whether this week's news materially updates that 1–4 quarter outlook."*
- Focus on ticker-specific, fundamentally relevant events; treat generic macro or trivia as noise.

RL Rubric (excerpt)

- `signal` $\in [-10, 10]$: net bullish/bearish update over 1–4 quarters.
- `risk_score` $\in [0, 10]$:
  - 0–2: almost no incremental downside.
  - 3–5: normal large-cap risk.
  - 6–8: clearly elevated risk.
  - 9–10: extreme, crisis-like.
- `confidence` $\in [0, 1]$: coverage quality and clarity.

Example: SEC Structural Prompt

SEC System Prompt (excerpt)

- *“You are a senior equity analyst synthesizing multiple chunk-level summaries of a 10-K / 10-Q filing into one consolidated SEC analysis.”*
- Emphasis on:
  - Business model: revenue streams, segments, geographies.
  - Competitive position: moat and industry structure.
  - Structural risks and exposures: balance sheet, legal/regulatory, concentration.
  - Long-term opportunities: secular growth, strategic shifts.
- Produces a small set of *top risks* with calibrated severities and aggregated flags for downstream RL use.
- Focus on a 1–3 year horizon and structural information, not quarter-to-quarter noise.

LLM Directional Signals: Cross-Channel Statistics

Statistic	SEC	EC	News	Technical	Senior
Count	6680	6680	6680	6680	6680
Mean	-1.10	3.36	0.98	-0.06	0.34
Std. deviation	2.48	1.37	3.76	5.91	2.71
Min	-6.00	-5.00	-9.00	-8.00	-8.00
25th percentile	-3.00	3.00	-2.00	-7.00	-2.00
50th percentile	-2.00	3.50	2.00	0.00	1.00
75th percentile	1.00	4.00	4.00	7.00	2.00
Max	5.00	5.00	8.00	8.00	7.00

- SEC: mildly negative, conservative structural risks.
- EC: strongly positive, low-variance management tone.
- News / Technical: near-zero mean but much higher variance.
- Senior: aggregated, moderated view across channels.

Recency of Fundamental Channels

Statistic	days_since_ec	days_since_sec
Count	6607	6484
Mean	45.37	44.74
Std. deviation	29.27	28.01
Min	0.00	0.00
25th percentile	22.00	21.00
50th percentile	44.00	43.00
75th percentile	66.00	66.00
Max	198.00	198.00

- EC/SEC updates arrive on a scale of weeks to months (median $\approx$ 6 weeks, max $\approx$ 6–7 months).
- They act as slow-moving fundamental baselines.

Training Procedure and Data Splits

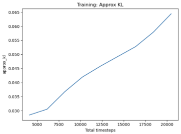
- **Start with price-only PPO**
  - Conv1D-based feature extractor over trading history.
  - 10 bps transaction cost.
  - No LLM signals: establish a baseline.
- **Hyperparameters tuned**
  - Learning rate (step size).
  - Total environment steps.
  - Rollout length between PPO updates.
- **Initial idea: fixed-start forward rolling**
  - Train on early window, validate on later window, roll forward.
  - Data scarcity at weekly frequency $\Rightarrow$ short windows, unstable estimates, underfitting.

Final Protocol

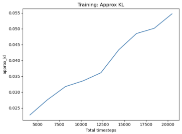
- Full horizon: 334 weekly decision steps.
- **Hyperparameter selection**
  - First 70% (weeks 1–234) as prefix.
  - Weeks 1–204: training.
  - Weeks 205–234: validation, averaged over 3 seeds.
- **Final training and test**
  - Retrain chosen configuration on weeks 1–234.
  - Evaluate once on held-out weeks 235–334.
- **For RQ1/RQ2/RQ3**
  - Reuse the same PPO hyperparameters and protocol.
  - Only change how LLM signals enter:
    - as observations (RQ1),
    - as risk-shaping terms (RQ2),
    - or via curriculum vs. all-in-one integration (RQ3).

Key Results

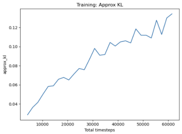
Training Convergence: Policy KL



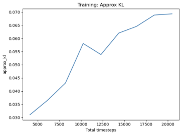
(a) Price-only PPO (best seed)



(b) RQ2: Tech risk penalty (best seed)



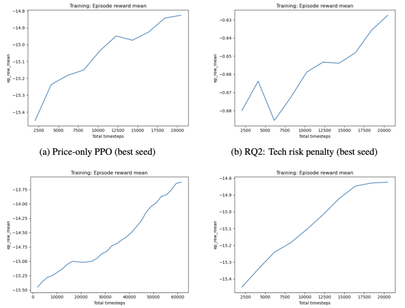
(c) RQ3: All-in-one, 60k steps (best seed)



(d) RQ3: Short→Long Stage 3 (seed 137)

- Approx. KL between old and new policies stays in a small band (≈ 0.03–0.07; up to ~ 0.12 for 60k all-in-one).
- Consistent with PPO/TRPO practice (trust-region-like updates).
- No large KL spikes $\Rightarrow$ no obvious optimisation collapse.

Training Convergence: Episode Reward



(a) Price-only PPO (best seed)

(b) RQ2: Tech risk penalty (best seed)

- Mean episodic rewards increase and then plateau for stable runs.
- Price-only PPO and RQ3 short→long show smooth growth.
- RQ2 + Tech risk is noisier but still shows net improvement.
- 60k all-in-one keeps improving as training steps increase.

Evaluation on Trading Performance

- Weekly portfolio log-returns:

$$r_{t+1} = \log\left(R^{\text{port}}_{t+1}\right),$$

  where $R^{\text{port}}_{t+1} = \frac{\mathrm{NAV}_{t+1}}{\mathrm{NAV}_t}$ is the one-week gross portfolio return after costs.
- NAV multiple (Net Asset Value):

$$\mathrm{NAV}_{\text{mult.}} = \frac{\mathrm{NAV}_T}{\mathrm{NAV}_0}.$$

- Annualised NAV (52-week year):

$$\mathrm{NAV}_{\text{ann.}} = \mathrm{NAV}_{\text{test}}^{\frac{52}{T}} - 1.$$

- Sharpe ratio (weekly):

$$\mathrm{Sharpe} = \frac{\mu_r}{\sigma_r},$$

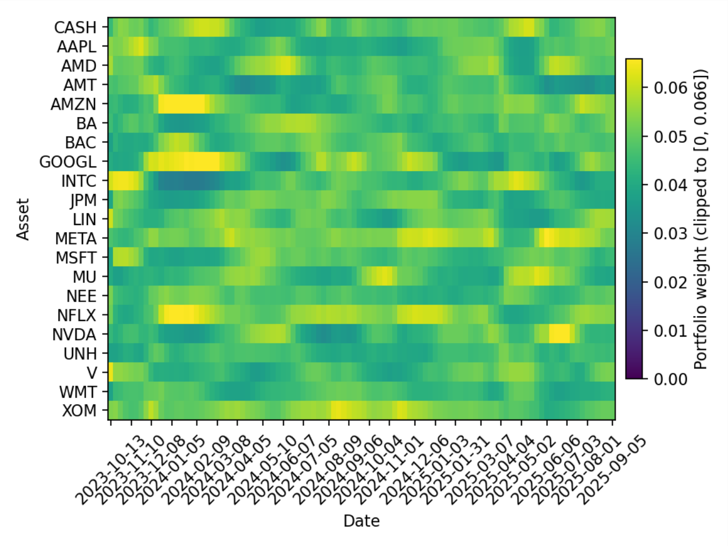
  where μ_r and σ_r are the sample mean and standard deviation of the weekly log-returns on the test window.
- Maximum drawdown (MDD) from the running peak:

$$M_t = \max_{0 \le s \le t} \mathrm{NAV}_s, \quad \mathrm{DD}_t = \frac{\mathrm{NAV}_t - M_t}{M_t}, \quad \mathrm{MDD} = \min_{0 \le t \le T} \mathrm{DD}_t.$$

Trading Performance (Oct. 2023 – Sep. 2025)

Regime	Strategy	NAV(mult.)	NAV(ann.)	Sharpe	MDD
Classical baselines (test window)					
Baseline	EW	**1.6539**	**1.2990**	**1.64**	**-0.16**
Baseline	MV	1.4573	1.2160	1.27	-0.20
Price-only PPO (test window backbone)					
PPO (price-only)	seed 42	**1.7267**	**1.3280**	**1.67**	**-0.16**
PPO (price-only)	seed 137	1.7191	1.3250	1.65	-0.17
PPO (price-only)	seed 256	1.6902	1.3140	1.61	**-0.16**
PPO (price-only)	mean (3 seeds)	1.7120	1.3228	1.64	-0.17
RQ1: LLM features as observations (3-seed means, test window)					
RQ1	PPO (price-only backbone)	**1.7120**	**1.3228**	**1.64**	-0.17
RQ1	+ SEC signal	1.6919	1.3130	1.60	-0.16
RQ1	+ EC signal	1.6807	1.3080	1.58	-0.16
RQ1	+ Senior signal	1.6747	1.3052	1.56	-0.16
RQ1	+ News signal	1.6665	1.3016	1.55	-0.16
RQ1	+ Technical-text signal	1.6515	1.2951	1.52	-0.17
RQ2: LLM-based risk shaping (best-Sharpe configs, test window)					
RQ2	PPO (price-only backbone)	1.7120	1.3228	1.64	-0.17
RQ2	+ EC risk penalty	1.6893	1.3134	1.62	**-0.16**
RQ2	+ News risk penalty	1.6868	1.3124	1.60	**-0.16**
RQ2	+ SEC risk penalty	1.6840	1.3113	1.60	**-0.16**
RQ2	+ Senior risk penalty	1.6711	1.3061	1.59	-0.16
RQ2	+ Technical-text risk penalty	**1.7013**	**1.3183**	**1.63**	-0.16
RQ3: Curriculum vs. all-in-one feature integration (3-seed means, test window)					
RQ3	*Curriculum: Long → Short (SEC+EC → +News → +Tech)*				
RQ3	Stage 1 (SEC+EC)	1.6770	1.3080	1.56	-0.17
RQ3	Stage 2 (SEC+EC+News)	1.6836	1.3110	1.57	-0.17
RQ3	Stage 3 (SEC+EC+News+Tech)	1.6528	1.2990	1.49	-0.17
RQ3	*Curriculum: Short → Long (+Tech → +News → +SEC+EC)*				
RQ3	Stage 1 (Tech)	1.6550	1.2990	1.52	-0.17
RQ3	Stage 2 (Tech+News)	1.6048	1.2790	1.40	-0.17
RQ3	Stage 3 (Tech+News+SEC+EC)	1.5462	1.2540	1.29	-0.18
RQ3	*All-in-one: full feature set from start*				
RQ3	All-in-one (20k steps)	1.6595	1.3010	1.54	-0.16
RQ3	All-in-one (60k steps)	**1.6854**	**1.3120**	**1.56**	**-0.16**

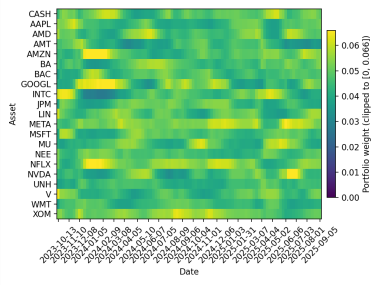
Price-Only PPO vs. Equal-Weight



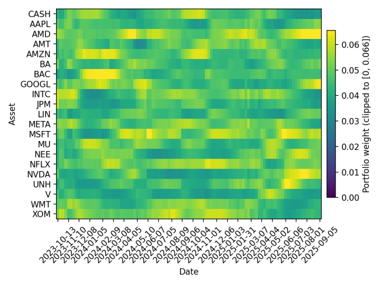
Why LLM Integration Does Not Beat Price-Only PPO

- **Performance level**
  - Price-only PPO already improves annualised NAV over EW/MV while keeping a similar weekly Sharpe.
  - LLM-based observation channels and risk penalties do not deliver a clear, robust improvement on top of this backbone.
- **Efficient market perspective**
  - Semi-strong EMH: public information (SEC, EC, news) is rapidly incorporated into prices, especially in deep US large-cap markets.
  - LLMs compress *already priced* information; any residual edge is likely small, noisy, and easily erased by transaction costs and regime shifts.
- **Role of LLM signals**
  - Act more like *behavioural regularisers* than robust alpha sources.

Price-Only vs. SEC-Augmented PPO (Heatmaps)



(a) Price-only PPO



(b) PPO + price + SEC signal

- SEC-augmented agent shows stronger tilts into AI-related names (e.g., AMD, NVDA, MSFT, GOOGL) than the price-only agent, after June 2025.
- Indicates that narrative information from filings is indeed influencing portfolio choices, even though overall Sharpe and NAV remain close to the price-only backbone.

Contributions, Limitations and Future Work

Contributions

- **End-to-end testbed**
  - Coupled a risk-aware weekly portfolio RL environment with a multi-genre LLM analysis pipeline.
  - Environment exposes structured observations, risk-aware rewards, transaction costs, and rich behavioural logs (turnover, HHI, cash usage, regime tags).
  - LLM side: unified pipeline that turns long, noisy text (SEC, EC, news, technical commentary) into a stable numeric interface for RL: `signal`, `risk_score`, `confidence`, `rationale`.
- **Systematic comparison of integration mechanisms**
  - RQ1: LLM features as observations.
  - RQ2: LLM-derived risk scores for reward shaping.
  - RQ3: curriculum-style vs. all-in-one feature schedules.
  - All evaluated on the same environment, data, and PPO backbone.
- **Empirical insights under strong baselines**
  - Benchmarked against EW, MV, and a strong price-only PPO.
  - On this weekly, large-cap US setting, price-only PPO is hard to beat: LLM

Limitations

- **Data and environment**
  - Single 20-ticker, S&P500-like universe on a weekly horizon; only a few hundred decision points.
  - Simplified market model: proportional transaction costs, basic cash asset, no explicit execution or liquidity modelling.
- **LLM pipeline**
  - Relies on hand-crafted prompts, schemas, and rubrics for each channel.
  - No end-to-end learning of the mapping from text to `signal`/`risk_score`.
  - Limited exploration of different LLM families or calibration strategies.
- **RL algorithms and baselines**
  - Focus on PPO with a targeted but limited hyper-parameter search.
  - No comparison to offline RL, imitation learning, or supervised allocation baselines.
  - Evaluation restricted to a single market and horizon.

Future Work

- **Scaling the environment**
  - Larger universes, multiple regions, and mixed frequencies.
  - Richer cost and risk models (liquidity-aware execution, leverage and constraint modelling).
- **Stronger LLM signal heads**
  - Supervised or RL-based calibration of `signal`/`risk_score` on historical outcomes.
  - Systematic comparison of different LLM families and prompt / fine-tuning strategies under a common rubric.
- **Broader RL and learning algorithms**
  - Compare PPO with alternative on-policy methods, value-based and actor–critic baselines, and offline RL.
  - Explore memory-augmented or meta-learning agents that adapt across regimes.
- **Beyond finance**
  - Reuse the "LLM analysts + risk-aware RL environment" pattern in other decision domains (operations, recommendation, logistics) where multi-genre text and numeric data meet.

Related Work and Positioning

RL for Trading and Portfolios

- Open-source frameworks:
  - FinRL, FinMeta: data pipelines, gym-style envs, RL baselines.
- Since ≈2017: deep RL agents for **portfolio allocation**
  - Agent outputs a weight vector over multiple assets.
  - Inputs: price/volume histories, sometimes technical indicators.
- Show that RL can learn dynamic rebalancing policies.
- **Limitation:** almost entirely numeric-only; text is not part of the state.

Textual Analysis and Financial LLMs

- **Classical textual analysis**
  - News, earnings calls, SEC filings contain predictive information about returns, volatility, and tail risk.
  - Traditional pipeline: dictionaries / simple classifiers $\rightarrow$ sentiment or tone scores $\rightarrow$ regressions.
- **Financial language models**
  - FinBERT: BERT adapted to financial news sentiment.
  - BloombergGPT, FinGPT: large models trained on financial corpora for analytics, robo-advising, sentiment-based trading, etc.
- Surveys highlight:
  - Trade-off: fine-tuned domain LMs vs. prompted general LLMs.
  - Need for *domain-grounded* evaluation beyond generic NLP tasks.
- This thesis: treat the LLM as an external analyst producing structured signals and risk scores for an RL agent.

Agentic LLM Systems in Finance

- Recent **agentic LLM** trend:
  - FinMem: layered memories and character profiling for stability.
  - TradingAgent, FinCon: multi-agent “virtual fund” with specialised roles (news, technicals, risk management) and verbal reflection to update beliefs.
- Common pattern:
  - Role-specialised agents around an LLM core.
  - Focus on news or single text channel, often at intraday scales.
- This thesis:
  - Light-weight two-tier analyst architecture (SEC, EC, News, Technical + senior analyst).
  - Outputs machine-readable signals for a weekly portfolio RL policy.

FinRL Competitions and Gap

- **FinRL competition trajectory**
  - Year 1: numeric-feature RL for single-stock intraday trading.
  - Year 2: RL with market feedback (LLM reads news, updated via RL).
  - Latest: RL agents consume LLM-generated signals from news.
- **Gap this thesis addresses**
  - Little work on *multi-genre* text (SEC, EC, news, technicals) feeding into a *risk-aware* portfolio RL agent.
  - Most prior setups are intraday; here we study a realistic weekly horizon with limited data.
  - Few studies analyse not just returns but the *behavioural profile* of the learned policy (turnover, concentration, regime behaviour).
- This thesis provides an end-to-end testbed to systematically compare:
  - LLM features as observations,
  - LLM-based risk shaping, and
  - curriculum vs. all-in-one integration.

Thank You

Thank you for your attention.

I'm happy to take questions and discuss.